Manuscript: PONE-D-26-50597

Candidate title: Source-only model selection and external evaluation of complete spectral-sharing pipelines in motor-imagery EEG decoding

Submitted title: Source-only model selection and limits of fixed spectral-sharing pipelines in cross-subject motor-imagery EEG decoding

This is an internal draft for author review. It has not been sent, and no permission to replace or revise the submitted files is assumed. Finalize the applicable editorial instruction and author approval before use.

Dear Editors,

I am preparing a technically reviewed revision candidate for manuscript PONE-D-26-50597, submitted to PLOS ONE on 9 October 2026. If you permit updated files, I would provide the candidate together with a change record and an issue-by-issue evidence matrix.

The revision clarifies two scientific aims: the reliability and participant heterogeneity of source-only training-duration selection, and the external limits of complete source-selected broad and fixed spectral-sharing pipelines. These comparisons differ in selected duration, input views, preprocessing, BatchNorm exposure and computation; they do not isolate the causal effect of spectral sharing. The adverse Cho2017 result and uncertain PhysioNet and Lee2019 results are retained. A separate BNCI sensor-level physiological characterization remains exploratory and does not establish the mechanism learned by a decoder.

The candidate also adds post hoc diagnostics reconstructed from archived CSP prediction tables, with aggregation and audit limits stated explicitly. These describe class-prediction asymmetry and stored probabilities; they do not involve target fitting, replace primary zero-calibration scores or establish why the pipeline transferred poorly. Follow-up source refitting and target-statistic interventions described in a separate audit report are not presented as independently validated results without their underlying records. No new model fit, prediction inference or physiology experiment was performed for this revision candidate.

Related work, terminology, citations and supplementary traceability have been revised. The frozen scientific results and the submitted v1.1 package are preserved, and the candidate includes a detailed change record. Original EEG remains available from the third-party providers; code, saved outputs and the evidence chain are publicly archived on GitHub. AI assistance is disclosed. There is no specific research funding and no competing interest.

I am the sole and corresponding author and remain responsible for the scientific content. Before providing any revised files, I will complete my review and approval of the candidate and follow your instructions. This letter does not request withdrawal or duplicate submission.

Sincerely,

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